MILITARY SPECIFICATION

DISPLAYS, DIODE, LIGHT EMITTING, SOLID STATE, RED, NUMERIC AND HEXADECIMAL, WITH ON BOARD DECODER/DRIVER

This specification is approved for use by the US Army Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for hermetically sealed red numeric and hexadecimal light emitting diode 4 X 7 dot matrix array displays. Two levels of product assurance are provided as specified in MIL-D-87157.

1.2 Part number. The complete part number shall be in accordance with MIL-D-87157.

1.2.1 Device type. Device type shall be as follows:

Device type	Function
01	Numeric indicator "0-9", a "-" sign, test pattern, and right hand decimal point.
02	Numeric indicator "0-9", a "-" sign, test pattern, and left hand decimal point.
03	Numeric indicator "±1" overrange display with right hand decimal point.
04	Hexadecimal indicator "0-9" and "A-F".

1.2.2 Device quality level. The device quality level shall be levels "A" and "B" as defined in MIL-D-87157.

1.2.3 Luminous intensity code. Luminous intensity code shall be as follows:

Luminous intensity code	Minimum luminous intensity µcd	Maximum luminous intensity µcd
C	44	79
D	59	106
E	79	142
F	106	190
G	142	255
H	190	342

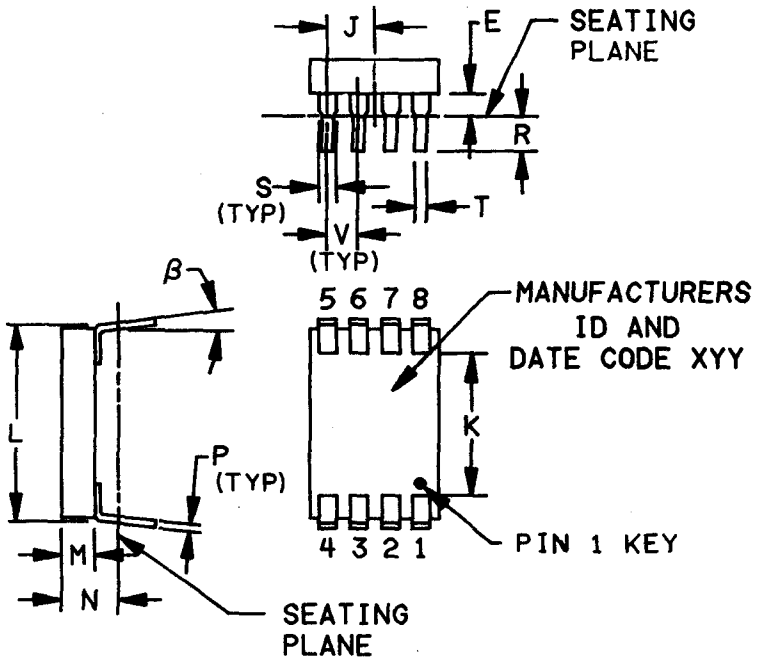
Test conditions shall be as specified in group A, subgroup 7.

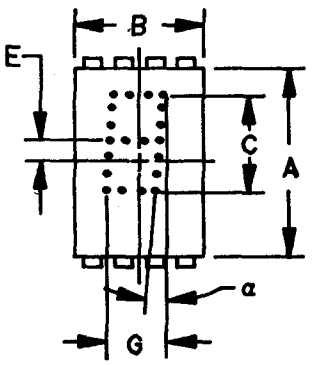
1.3 Physical dimensions. See figure 1.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Electronics Research and Development Command, ATTN: DELET-R-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

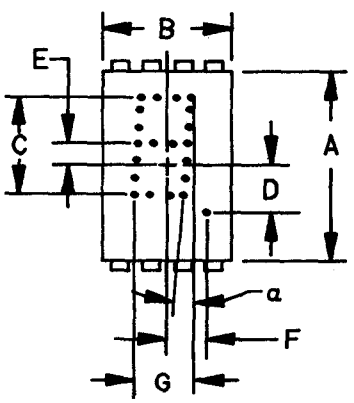
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SYM	DIMENSIONS			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.515	.545	13.12	13.88
B		.400		10.20
C	.275	.305	7.02	7.78
D	.205	.235	5.22	5.98
E	.045	.075	1.12	1.88
F	.105	.135	2.62	3.38
G	.175	.205	4.42	5.18
H	.125	.155	3.22	3.98
I	.060	.090	1.52	2.28
J	.135	.165	3.42	4.18
K	.385	.415	9.82	10.58
L	.585	.815	14.82	15.58
M	.095	.125	2.42	3.18
N	.155	.185	3.92	4.68
P	.009	.015	.22	.38
R	.120	.150	3.02	3.78
S	.035	.065	.92	1.68
T	.017	.023	.42	.58
V	.095	.105	2.37	2.63
α	4°	6°	4°	6°
β	0°	10°	0°	10°

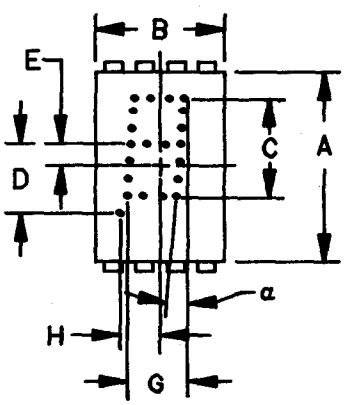




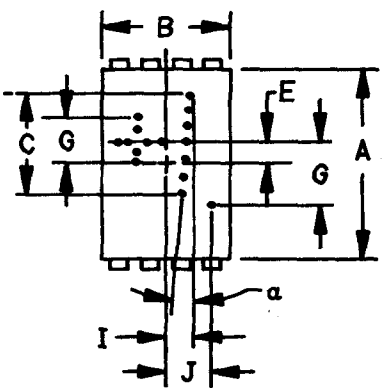
Device type 04



Device type 01



Device type 02



Device type 03

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Physical dimensions.

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1.4 Absolute maximum ratings.

Maximum total power dissipation (see 6.3):	
Device types 01, 02, and 04 1/	935 mW
Device type 03 2/	320 mW
Logic supply voltage (V_{CC})	7.0 V dc
Voltage applied to input logic pins (see 6.3)	7.0 V dc
Voltage applied to blanking input (see 6.3)	
Device type 04	V_{CC}
Maximum solder temperature 3/	260°C
Ambient operating temperature 4/	-55°C to +100°C
Ambient storage temperature	-65°C to +125°C

1.5 Recommended operating conditions.

Logic supply voltage	4.5 to 5.5 V dc
Operating temperature range 5/	-55°C to +100°C
Enable pulse width 6/	100 ns minimum
Set up time 6/ 7/	50 ns minimum
Hold time 6/ 8/	50 ns minimum
Enable pulse rise time 6/	200 ns maximum

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

1/ V_{CC} = 5.5 V dc (numeral 5 lighted for all types. Decimal point lighted on types 01 and 02 only).

2/ I_F = 10 mA dc per LED (see 6.3). Current limiting resistors must be used.

3/ t $\leq$5 seconds at .062 inch below seating plane.

4/ $R_{\theta CA}$ = 35°C/W maximum.

5/ $R_{\theta CA}$ = 35°C/W maximum at +100°C.

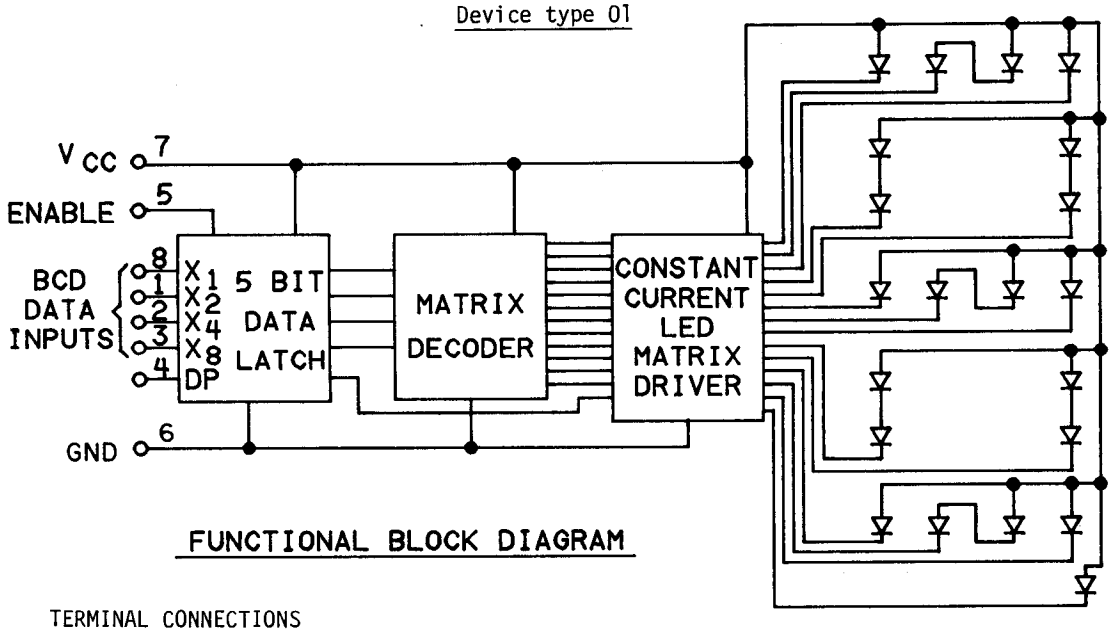
6/ See figure 2 for timing diagrams of series logic.

7/ Minimum set up time is the interval immediately preceding the positive going transition of the enable input during which interval the data to be displayed must be maintained at the latch data inputs to ensure its recognition.

8/ Minimum hold time is the interval immediately following the positive going transition of the enable input during which interval the data to be displayed must be maintained at the latch data inputs to ensure its continued recognition.

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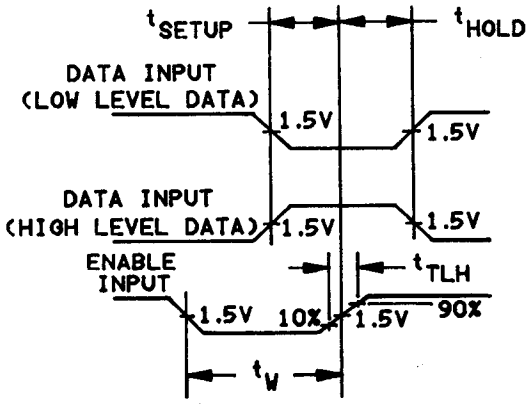
Device type 01



FUNCTIONAL BLOCK DIAGRAM

TERMINAL CONNECTIONS

PIN	DEVICE 01
1	Input 2
2	Input 4
3	Input 8
4	Decimal point
5	Latch enable
6	Ground
7	V_{CC}
8	Input 1



TIMING DIAGRAM

TRUTH TABLE				
BCD DATA				device 01
X_8	X_4	X_2	X_1	
L	L	L	L	0
L	L	L	H	1
L	L	H	L	2
L	L	H	H	3
L	H	L	L	4
L	H	L	H	5
L	H	H	L	6
L	H	H	H	7
H	L	L	L	8
H	L	L	H	9
H	L	H	L	8
H	L	H	H	(BLANK)
H	H	L	L	(BLANK)
H	H	L	H	---
H	H	H	L	(BLANK)
H	H	H	H	(BLANK)

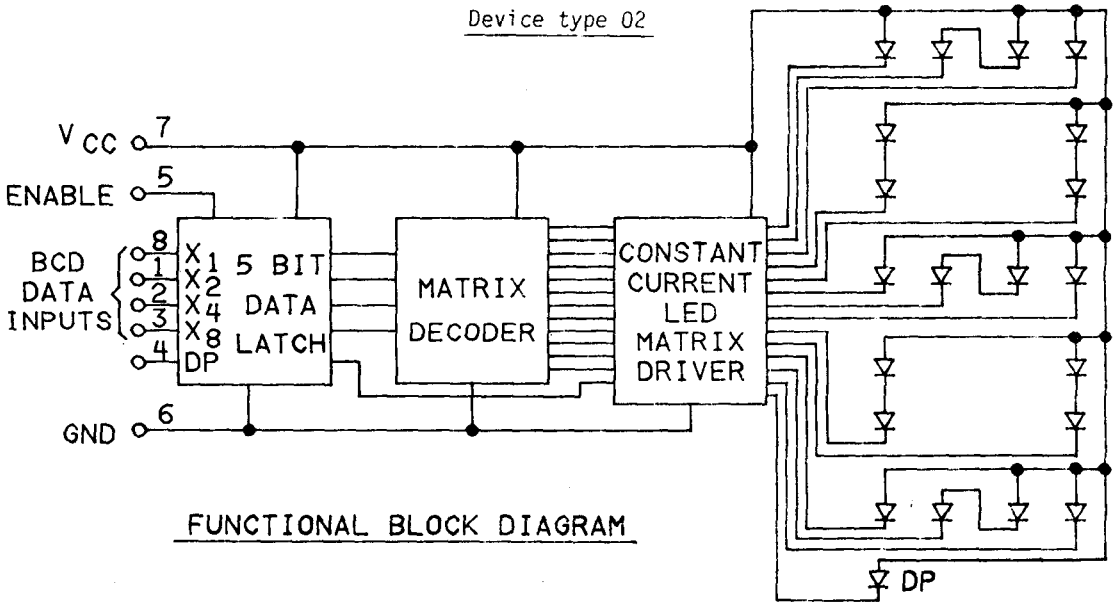
DECIMAL PT	ON	V_{DP} - L
	OFF	V_{DP} - H
ENABLE	LOAD DATA	V_E - L
	LATCH DATA	V_E - H

NOTE: L = Logic low
H = Logic high, with the enable input at logic high, changes in BCD inputs or D.P. input have no effect upon display memory, display character or D.P.

FIGURE 2. Functional block diagrams, terminal connections, timing diagrams, and truth tables.

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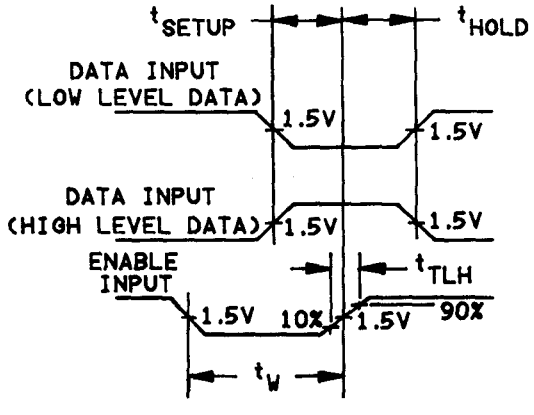
Device type 02



FUNCTIONAL BLOCK DIAGRAM

TERMINAL CONNECTIONS

PIN	DEVICE 02
1	Input 2
2	Input 4
3	Input 8
4	Decimal point
5	Latch enable
6	Ground
7	V_{CC}
8	Input 1



TIMING DIAGRAM

TRUTH TABLE				
BCD DATA				device 02
X_8	X_4	X_2	X_1	
L	L	L	L	0
L	L	L	H	1
L	L	H	L	2
L	L	H	H	3
L	H	L	L	4
L	H	L	H	5
L	H	H	L	6
L	H	H	H	7
H	L	L	L	8
H	L	L	H	9
H	L	H	L	8
H	L	H	H	(BLANK)
H	H	L	L	(BLANK)
H	H	L	H	---
H	H	H	L	(BLANK)
H	H	H	H	(BLANK)

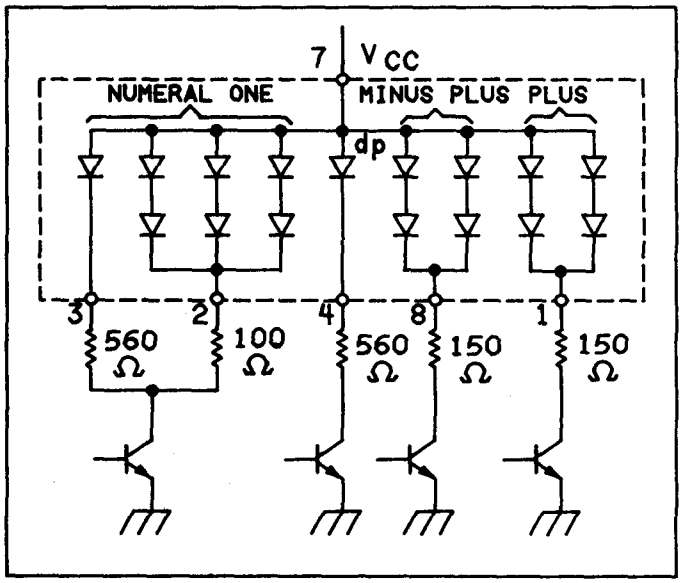
DECIMAL PT	ON	V_{DP} - L
	OFF	V_{DP} - H
ENABLE	LOAD DATA	V_E - L
	LATCH DATA	V_E - H

NOTE: L = Logic low
H = Logic high, with the enable input at logic high, changes in BCD inputs or D.P. input have no effect upon display memory, display character or D.P.

FIGURE 2. Functional block diagrams, terminal connections, timing diagrams, and truth tables - Continued.

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Device type 03



TYPICAL DRIVING CIRCUIT; V_{CC} = 5.0 VOLTS

Terminal connections.

Pin	Device 03
1	Plus
2	Numeral One
3	Numeral One
4	DP
5	Open
6	Open
7	V_{CC}
8	Minus/Plus

TRUTH TABLE

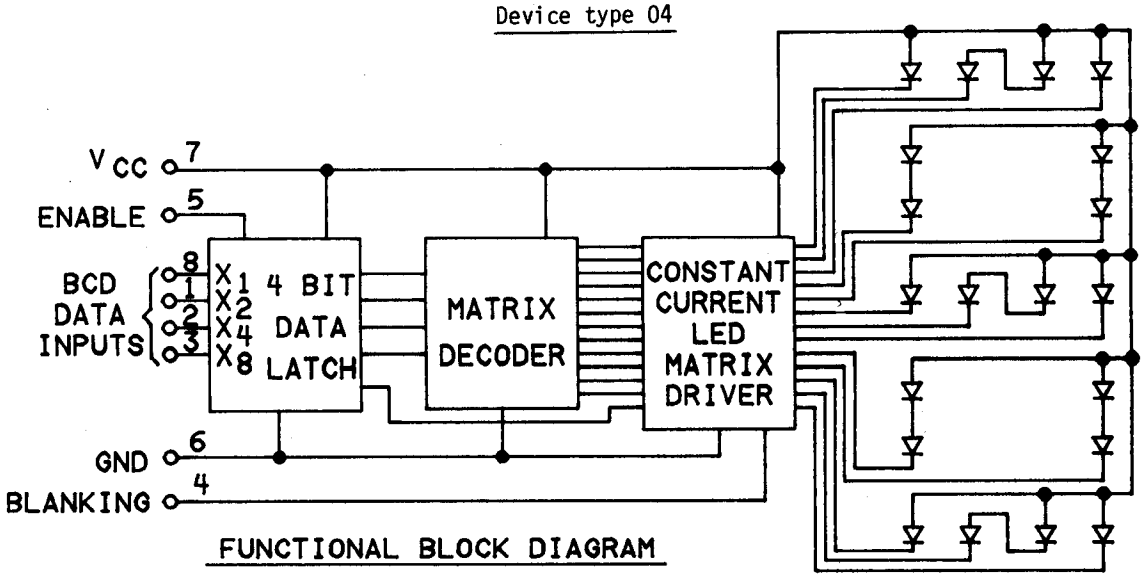
CHARACTER	PIN			
	1	2,3	4	8
+	H	X	X	H
-	L	X	X	H
1	X	H	X	X
Decimal Point	X	X	H	X
Blank	L	L	L	L

NOTES: L = LED driver transistor - OFF
H = LED driver transistor - ON
X = Don't care

FIGURE 2. Functional block diagrams, terminal connections, timing diagrams, and truth tables - Continued.

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Device type 04



FUNCTIONAL BLOCK DIAGRAM

Terminal connections.

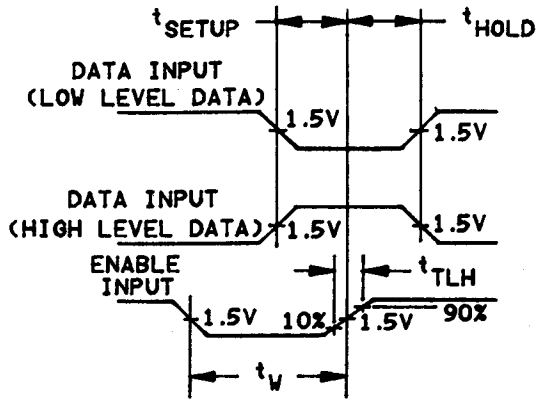
Pin	Device 04
1	Input 2
2	Input 4
3	Input 8
4	Blanking control
5	Latch enable
6	Ground
7	V_{CC}
8	Input 1

NOTE: L = Logic low
H = Logic high

1/ The blanking input controls the display LED drivers, but does not have an affect upon the display memory.

2/ With the enable high, changes in BCD inputs have no effect upon display memory or display character.

TRUTH TABLE				
BCD DATA				device 04
X_8	X_4	X_2	X_1	
L	L	L	L	0
L	L	L	H	1
L	L	H	L	2
L	L	H	H	3
L	H	L	L	4
L	H	L	H	5
L	H	H	L	6
L	H	H	H	7
H	L	L	L	8
H	L	L	H	9
H	L	H	L	A
H	L	H	H	B
H	H	L	L	C
H	H	L	H	D
H	H	H	L	E
H	H	H	H	F
BLANKING 1/				DISPLAY ON V_B - L
				DISPLAY OFF V_B - H
ENABLE 2/				LOAD DATA V_E - L
				LATCH DATA V_E - H



TIMING DIAGRAM

FIGURE 2. Functional block diagrams, terminal connections, timing diagrams, and truth tables - Continued.

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SPECIFICATIONS

MILITARY

MIL-D-87157 - Displays, Diode, Light Emitting, Solid State, General Specification for.
MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-D-87157 and as specified herein.

3.2 Product assurance requirements. The product assurance program and manufacturing certification requirements shall be in accordance with appendix D of MIL-S-19500.

3.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-D-87157 and as follows:

I_{BH} - Blanking input current "1" state

I_{BL} - Blanking input current "0" state

I_{EH} - Enable high level input current

I_{EL} - Enable low level input current

I_{IH} - Logic data input current "1" state

I_{IL} Logic data input current "0" state

t_{hold} - Time data must be held after positive transition of enable input.

t_{setup} - Time data must be held before positive transition of enable input.

t_{TLH} - Enable input pulse rise time

V_B - Voltage applied to blanking input

V_{BH} - Blanking high voltage; display blanked

V_{BL} - Blanking low voltage; display not blanked

V_{DP} - Voltage applied to decimal point

V_E - Voltage applied to enable input

V_{EH} - Enable input high voltage; data not being entered

V_{EL} - Enable input low voltage; data being entered

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η_V	-	Luminous efficacy $\frac{\text{luminous power}}{\text{radiant power}}$	= 683 lumens/watt at 555 nm.
V_I	-	Voltage applied to input logic	
V_{IH}	-	Logic data input voltage "1" state	
V_{IL}	-	Logic data input voltage "0" state	
λ_{PEAK}	-	Peak wavelength	
λ_d	-	Dominant wavelength	

3.4 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-D-87157 and as shown on figure 1. Although eutectic die bonding is preferred, epoxy die bonding may be performed. When epoxy die bonding is used, the material, cure cycle, and other pertinent data shall be approved by the qualifying activity.

3.4.1 Terminal connections. The terminal connections and functions shall be as specified on figure 2.

3.4.2 Truth tables. The truth tables and resultant displays of the binary data in the latches shall be as specified on figure 2.

3.5 Lead material and finish.

3.5.1 Lead material. The lead material shall be in accordance with MIL-D-87157.

3.5.2 Lead finish. The lead finish shall be in accordance with MIL-D-87157.

3.6 Marking. Marking shall be in accordance with MIL-D-87157 and as specified herein. At the option of the manufacturer, the marking may be on more than one line as shown by the following example:

M87157/
00101ACC

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-D-87157 and as specified herein.

4.2 Screening (level A only). Screening shall be in accordance with table I of MIL-D-87157 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table I of MIL-D-87157)	Conditions
1.	See 4.2.1
2.	24 hours minimum at T_A = 125°C
3.	Condition B (10 cycles) $t_{extreme} \geq 10$ minutes. Transfer time ≤ 5 minutes.
4.	10,000 G Y1, for 1 minute minimum
5.	Test condition G or H maximum leak rate = 5×10^{-8} atm cm^3/s
6.	Test condition A, C, E or F
7.	Subgroups 1 and 7 of table I herein
8.	160 hours at T_A = 100°C. See 4.2.2 and figure 3.
9.	Subgroups 1 and 7 of table I herein
10.	ΔI_V = -20% of group A value; ΔI_{IH1} = ±10 µA dc; ΔI_{CC1} = ±10 mA dc; ΔI_{EH1} = ±13 µA dc
11.	See 4.2.3

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4.2.1 Internal visual (precap) inspection. Internal visual (precap) inspection shall be in accordance with method 2072 of MIL-STD-750, except crack reject criteria shall be as follows:

a. Any crack which extends into an active region or through a guard ring.

b. Any crack that exceeds 1.0 mil in length inside the scribe grid or scribe line that points toward operating metallization or functional elements.

c. Any crack that exceeds 5.0 mils in length.

4.2.2 Burn-in. The burn-in test circuits shall be as specified on figure 3. All displays shall be cooled in accordance with 4.5.1.

4.2.3 Visual and mechanical inspection. Displays shall be examined under a magnification between 3X and 10X with a field of view sufficiently large to contain the entire display. Displays that exhibit any of the following criteria shall be rejected:

a. Lead identification, markings (content, placement, and legibility), materials, construction, and workmanship not in accordance with MIL-D-87157.

b. Defects or damage resulting from manufacturing, handling, or testing.

c. Visible evidence of corrosion, contamination, or breakage (grossly bent or broken leads, cracked seals except for glass meniscus), and peeled, flaked, or blistered plating. (Discoloration of the finish shall not be cause for failure unless there is evidence of flaking, pitting, or corrosion).

d. Leads that are not intact and alined in their normal location, free of sharp or unspecified lead bends and (for ribbon leads) free of twist outside the normal lead plane.

e. Leads that are not free of foreign material such as paint, or other adherent deposits, or dust.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-D-87157. The following additional tests shall be required for qualification approval.

4.3.1 Off-axis luminous intensity. Off-axis luminous intensity at 40 degrees from the normal shall be measured in accordance with 4.5.5. Two samples shall be required and the minimum acceptable value of off-axis luminous intensity shall be 50 percent of the on-axis luminous intensity.

4.3.2 Dimming range capability. This test shall demonstrate dimming range capability. Two samples shall be required and the dimming range shall be 1000:1. Dimming range shall be measured in accordance with 4.5.6.

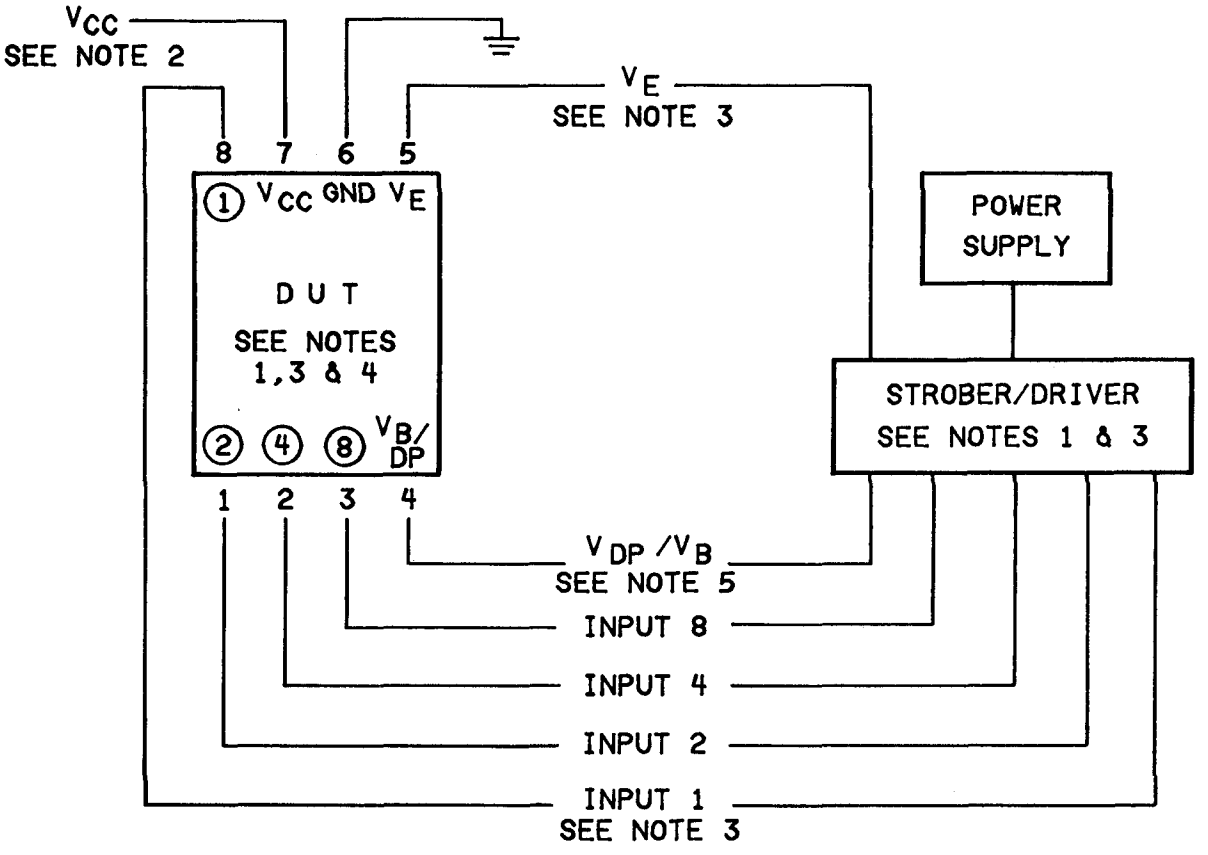
4.3.3 Display color and color uniformity. The purpose of this test is to determine that the color of the display falls visually within the required color range. Two displays shall be measured in accordance with 4.5.7 to determine dominant wavelength. Measurements must fall between 625 and 660 nanometers for acceptable units.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-D-87157 and as specified herein. If the manufacturer chooses the following option for testing, the sample displays that are to be used in group C inspection shall be designated as such prior to conducting the referenced group B test. The number of failed displays to be counted for lot acceptance or rejection as a result of the group C test shall be equal to all failed displays of the test in group B inspection which were predesignated for use in group C inspection, plus any additional failures occurring during group C testing.

a. For the steady state operation life test in group C inspection, the manufacturer shall have the option of using all or a portion of the sample already subjected to 340 hours of group B life testing for an additional 660 hours of testing to meet the 1000 hour requirement.

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Device types 01, 02, and 04.

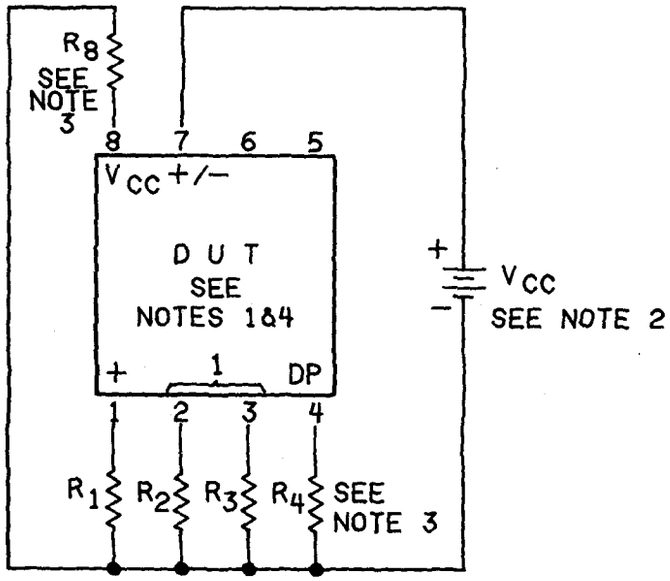


NOTES:
1. Strober/driver cycles D.U.T. through logic states at 1 character/second rate.
2. V_{CC} = 5.0 V.
3. Logic and enable input allowed voltage levels: $V_{IH} = V_{EH}$ = 2.4 V to V_{CC}; $V_{IL} = V_{EL}$ = 0 V to 0.4 V.
4. T_A = +100°C.
5. V_{DP} for device types 01 and 02; V_B for device type 04.

FIGURE 3. Burn-in circuits.

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Device type 03



NOTES:
1. 8 mA average/LED; nominal current values are: pin 1 and pin 8 = 16 mA, pin 2 = 24 mA, pin 3 and pin 4 = 8 mA.
2. V_{CC} = 5.0 V.
3. Current limiting resistors; 2%, 1/4 W: $R_1 = R_8 = 24\Omega$, $R_2 = 15\Omega$, $R_3 = R_4 = 91\Omega$.
4. $T_A = +100°C$.

FIGURE 3. Burn-in circuits - Continued.

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b. Group C inspection shall be performed when the first lot is produced and thereafter every 6 months during production.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-D-87157 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table III of MIL-D-87157 and table II herein.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IV of MIL-D-87157 and table III herein.

4.4.4 Inspection of packaging. Inspection of packaging shall be in accordance with MIL-D-87157.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Life test and burn-in cooldown procedure. The displays shall be cooled to room temperature prior to the removal of bias. The interruption of bias for up to 1 minute for the purpose of moving the displays to cooldown positions separate from the chamber within which life testing or burn-in was performed shall not be considered removal of bias. Alternately, the bias may be removed during cooling provided the case temperature of displays under test is reduced to a maximum of 35°C within 30 minutes after the removal of the test conditions.

4.5.2 Lead integrity. Each lead of each test display shall be bent inward through an angle sufficient to cause the lead to retain a permanent bend (after stress removal) of at least 15 degrees measured at the lead extremities. At the completion of the initial bend, the leads shall be returned to their approximate original position. This procedure shall be repeated for a total of three complete bends.

4.5.2.1 Failure criteria. After removal of the stress, any evidence of breakage, loosening, or relative motion between the lead (terminal) and the display body shall be considered a failure. Meniscus cracks shall not be cause for rejection.

4.5.3 Moisture resistance. Moisture resistance shall be conducted in accordance with method 1021 of MIL-STD-750 and 4.5.3.1, 4.5.3.2, and 4.5.3.3 herein.

4.5.3.1 Failure criteria. Any display with illegible markings or which, when using a magnification of 10X to 20X, shows evidence of corrosion, flaking, or pitting of the finish that will interfere with the application of the device, shall be considered a failure. The finish shall include the entire exposed lead area from the meniscus to the tip and all other exposed metal surfaces. Markings shall include all specified markings.

4.5.3.2 Finish. When viewed at 10X to 20X magnification, any lead breakage, deep penetration of metal elements (penetration either over an area or at a single point which is greater than 50 percent of the diameter, width, or thickness), peeling or blistering which indicates loss of adhesion of the finish to the base metal or any significant formation of corrosion products (especially any bridging between unconnected metal elements by products of corrosion) shall constitute a failure.

4.5.3.3 Marking. Any specified markings which are missing in whole or in part, faded, smeared, blurred, or shifted (dislodged) to the extent that they cannot be readily identified from a distance of at least 6 inches with normal room lighting and without the aid of magnification, or with a viewer having a magnification no greater than 3X, shall constitute a failure.

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4.5.4 Axial luminous intensity.

4.5.4.1 Apparatus. The equipment for this measurement shall consist of a photometer designed to respond to incident luminous flux density. The output of the photometer shall be linearly related to luminous incidence over the range of levels encountered in calibration and measurement. The output may be a voltage or a current or may be rendered directly in the units of luminous incidence.

4.5.4.1.2 Spectral response. The relative response of the photometer shall be within ±6 percent of the photopic luminous efficacy as defined by the American National Standards Institute PR-16-1980 for all wavelengths within the spectrum of the device to be measured.

4.5.4.1.3 Receptance pattern. The off-axis receptance of the photometer shall be constant over a large enough angle so that it responds equally to light from all parts of the display to be measured. An effective plane of receptance (image of the detecting surface) shall be defined with respect to which the calibration can be performed.

4.5.4.2 Calibration of photometer (see figure 4). The photometer shall be calibrated with a National Bureau of Standards (NBS) Detector Response Transfer Intercomparison Package (DRIP), an OSRAM FCR 6333 100W, 12V, 3300°K quartz tungsten halogen incandescent lamp or equivalent, and Omega Optical full blocking 10 nanometer bandwidth interference filters or equivalent. Peak transmission wavelengths shall be centered within the 550 nm to 740 nm range. The correlation factors derived by comparing the response of the photometer with the calculated photometric response of the NBS DRIP unit shall be used as the photometer calibration.

4.5.4.2.1 Calibration procedure. The equipment shall be mounted on an optical bench and a photo detector reference plane established. A baffle box shall be used to collimate the light and support the interference filters. The exit aperture shall be placed as close to the photo detectors as possible and shall form a spot of light of known area (A_o) that underfills the active areas of the photometer and DRIP unit photo detectors. The distance between the quartz lamp filament and the reference plane shall be within 31.4 inch (800 mm) and 35.4 inch (900 mm).

4.5.4.2.2 Equipment calibration measurements and calculation. The quartz lamp shall be illuminated and one hour shall be allowed for stabilization. The photocurrent output in amperes of the NBS DRIP unit and the luminous incidence by the photometer in lumens per square meter shall be measured at the reference plane for each interference filter used. From the responsivity table supplied with the NBS DRIP unit, the DRIP responsivity at the peak wavelength shall be determined for each interference filter used. The radiometric incidence of the NBS DRIP unit at the peak wavelength of each interference filter shall be calculated as follows:

$$E_{e_o}(\lambda_p) = \frac{I_{p_o}(\lambda_p)}{A_o \cdot R_o(\lambda_p)}$$

where $E_{e_o}(\lambda_p)$ = Radiometric incidence in watts/m^2 at peak wavelength of interference filter.

I_{p_o} = Photocurrent in amperes at peak wavelength of interference filter.

A_o = Illuminated active area of the detectors used during calibration, (m^2).

R_o = Responsivity in amperes per watt of the NBS DRIP unit at the peak wavelength of the specified interference filter.

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Luminous efficacy in lumens per watt for each interference filter used shall be calculated as follows using 1 nanometer intervals across the filter transmission spectrum:

$$\eta_v = \frac{683 \int T_{FILTER}(\lambda)\, \bar{y}(\lambda)\, d\lambda}{\int T_{FILTER}(\lambda)\, d\lambda}$$

where η_v = luminous efficacy in lumens/watt.

T_λ = relative spectral transmission of an interference filter.

$\bar{y}_\lambda$ = 1931 luminosity function (CIE photopic curve).

The photometer correlation factor at the peak wavelength for each interference filter shall be calculated as follows:

$$K(\lambda p) = \frac{E_v(\lambda_p - \text{measured})}{E_{e_o}(\lambda_p) \cdot \eta_v}$$

where $K_{(\lambda_p)}$ = photometer correlation factor at the peak wavelength for each interference filter.

$E_v(\lambda_p - \text{measured})$ = measured luminous incidence by the photometer in lumens/m^2.

$E_{e_o}(\lambda_p) \cdot \eta_v$ = calculated photometric response of the NBS DRIP unit.

The values of $K_{(\lambda_p)}$ vs λ_p shall be plotted graphically and the correlation value for 655 nanometers (K_{655}) shall be determined.

4.5.4.3 Operation of photometer. The photometer shall be adjusted to measure on axis light output as luminous intensity in candelas. The correct value for luminous intensity shall be calculated by using the following equation:

$$I_v = \frac{1}{K_{655}} \cdot \text{Photometer reading}$$

where I_v = corrected value of photometer output in candelas.

K_{655} = photometer correlation factor at 655 nanometers.

4.5.5 Luminous intensity off-axis (see figure 5). The display shall be rotated relative to the photometer detector head or the position and angle of the detector head can be varied provided that the distance from the display digit remains constant. The character "8" shall be illuminated on devices 01, 02, and 04. The character "+1" shall be illuminated on device 03. Measurement of luminous intensity ($I_{v\theta}$) at the specified angle θ from the normal axis shall be made from four directions X', X", Y', and Y" as shown on figure 5.

4.5.6 Dimming range. Dimming range capability shall be determined by first measuring the on-axis luminous intensity of the display under test while the display is being driven at the standard dc drive conditions. The display shall then be dimmed by the appropriate pulse width modulation (see 4.5.6.1), and the luminous intensity of the display shall again be measured. The dimming range shall be determined by the ratio of on-axis luminous intensity at normal drive conditions to the on-axis luminous intensity at pulse width modulated drive conditions.

4.5.6.1 Pulse width modulation procedure. Display dimming shall be accomplished by pulse width modulation using the waveform shown on figure 6. The display under test is blanked when the pulse modulated voltage V_{PWM} is less than 0.8 V and is turned on when the pulse modulated voltage is greater than 3.5 V. Dimming circuitry for displays covered by this specification shall be as indicated on figure 7. Duty factor period time shall be equal to 10 milliseconds and the display on time shall be adjusted to achieve the dimming range specified in 4.5.6. The displayed character shall be legible in a dark environment over the dimming range specified in 4.5.6.

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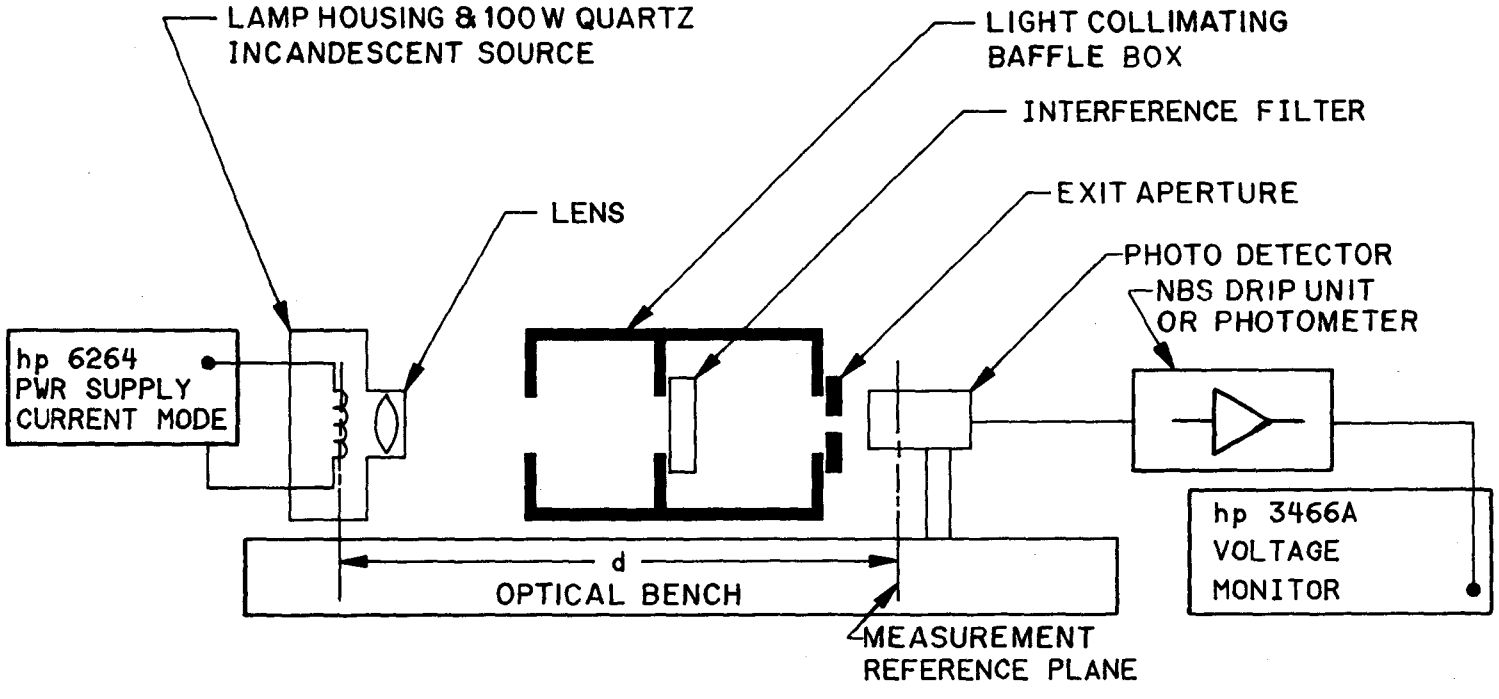


FIGURE 4. Photometer calibration block diagram.

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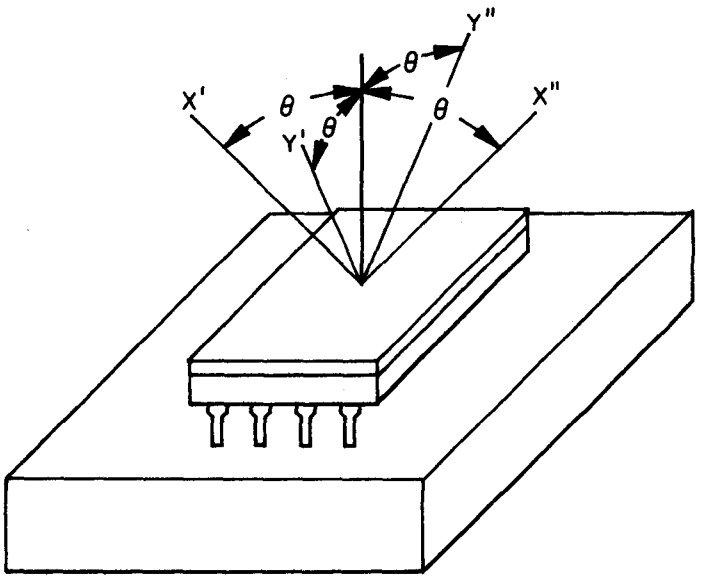
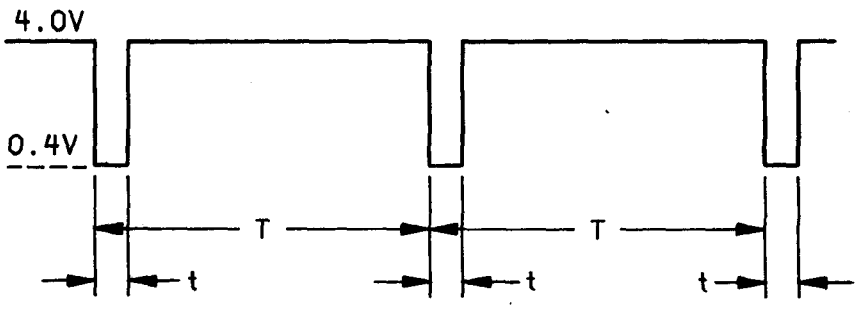


FIGURE 5. Off-axis luminous intensity.



T = 10 MILLISECONDS
t = 5 MICROSECONDS

FIGURE 6. Pulse width modulation waveform.

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Device types 01 and 02.

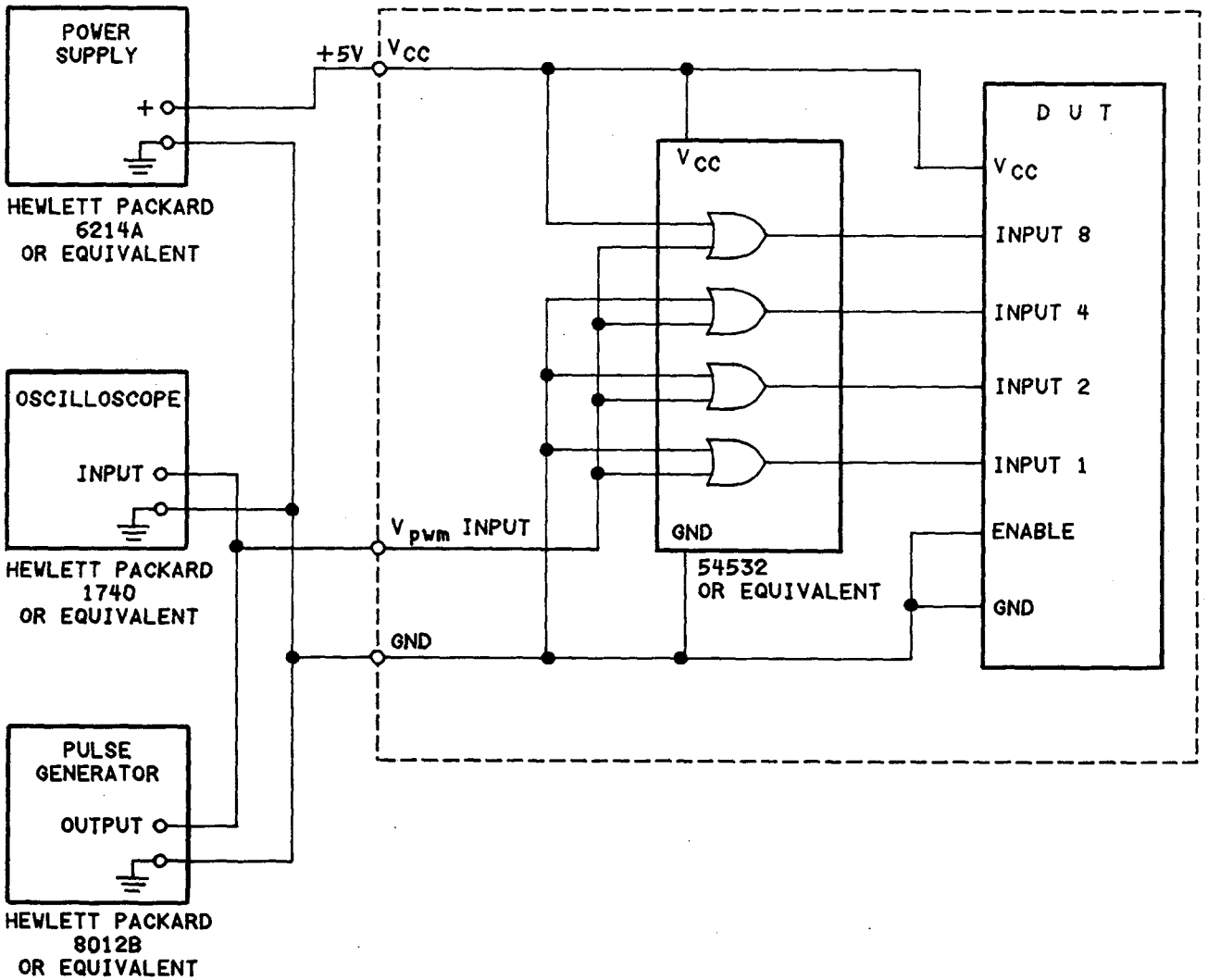


FIGURE 7. Dimming circuit.

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Device type 03

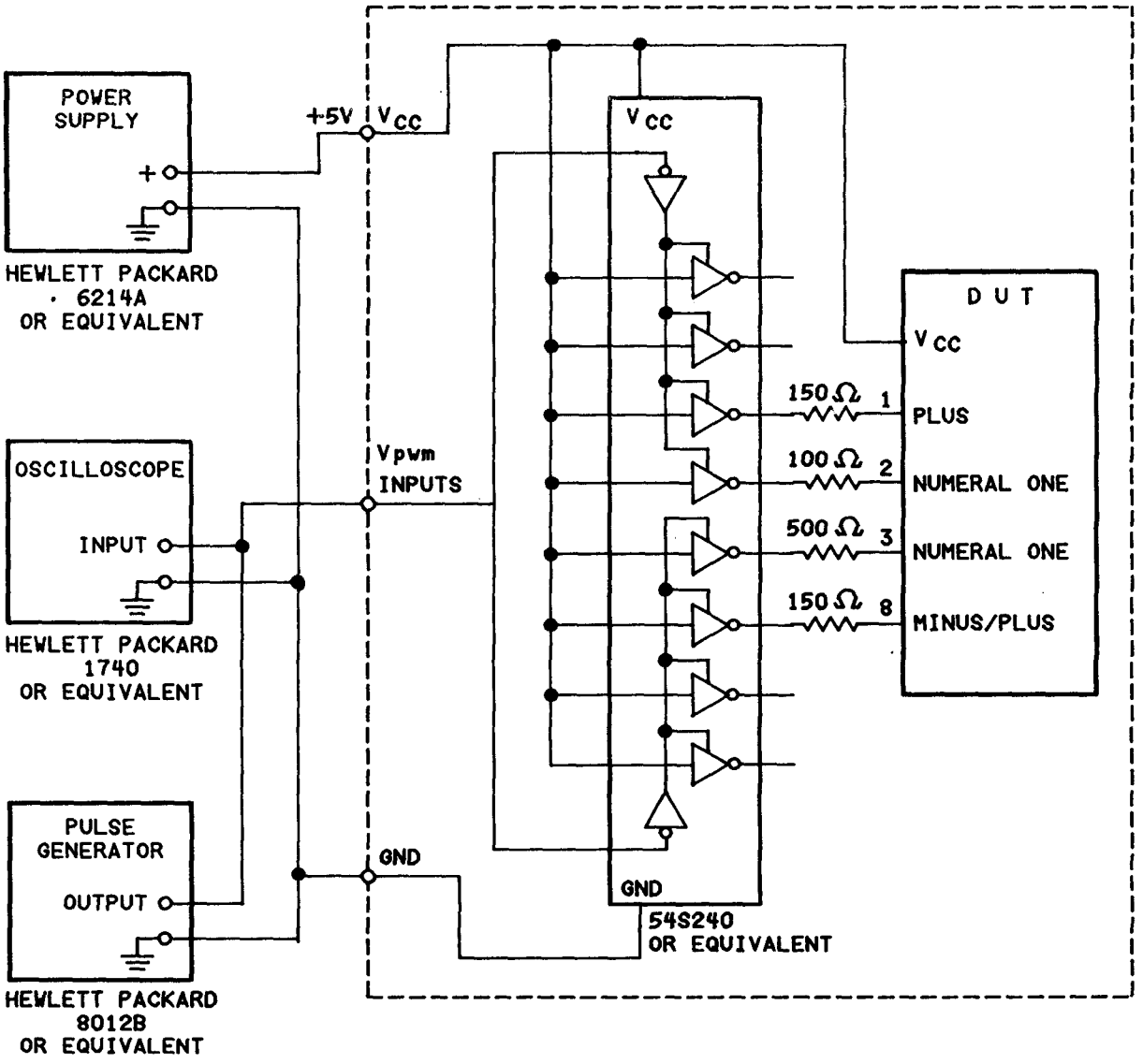


FIGURE 7. Dimming circuit - Continued.

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Device type 04

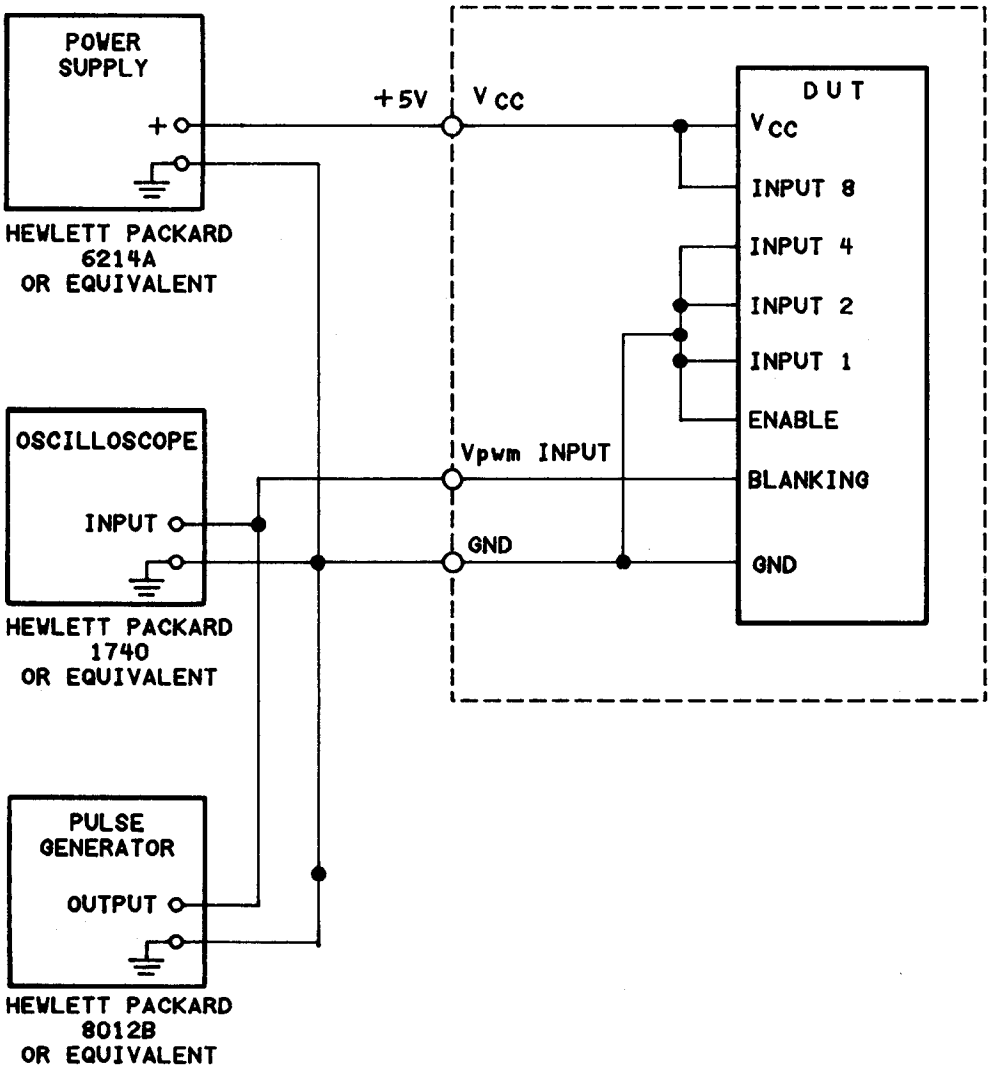


FIGURE 7. Dimming circuit - Continued.

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4.5.7 Display color, dominant wavelength.

4.5.7.1 Apparatus. A spectroradiometer, designed to measure the spectral distribution of light sources, shall be used to measure the spectral distribution of the display under test. The spectroradiometer shall have a spectral bandwidth of 5 nanometers or less, a sensitivity of 2.4 x 10^{-9} watts/steradian per nanometer at 655 nanometers and a flat relative response within ±5 percent from 550 to 730 nanometers.

4.5.7.2 Calibration. Calibration of the spectroradiometer shall be accomplished by using a light source of known spectral output and dividing the reading of the spectroradiometer at each wavelength by the output of the light source at that wavelength. By generating a table of weighting factors at each wavelength of the spectroradiometer, a uniform reading at all wavelengths for a light source of uniform output may be obtained by multiplying the reading of the spectroradiometer at each wavelength by the appropriate weighting factor.

4.5.7.3 Procedure. The LED display to be measured shall be mounted perpendicular to the entrance aperture of the spectroradiometer at a distance such that the flux emitted by all of the light emitting diodes is detected. The display shall be spectrally scanned between the wavelengths of 550 and 740 nanometers and the resulting spectrum normalized to 1.00 at the peak wavelength. The 1931 CIE x, y color coordinates may be derived by first integrating the normalized LED display spectrum to obtain the two tristimulus values X, Y and then calculating the 1931 chromaticity coordinates as follows:

$$X = \int_{550}^{740} f(\lambda)\ \bar{x}_\lambda\ d_\lambda$$

$$Y = \int_{550}^{740} f(\lambda)\ \bar{y}_\lambda\ d_\lambda$$

where $\bar{x}_\lambda$ and $\bar{y}_\lambda$ are the 1931 CIE color matching functions
and f_λ is the normalized radiated spectrum.

The integration shall be performed at a maximum of 5 nanometer intervals to determine color coordinates. The 1931 chromaticity coordinates for light emitting diodes may be calculated directly from the values of X and Y as determined above, using the following relationships:

$$x = \frac{X}{X + Y} \quad ; \quad y = \frac{Y}{X + Y}$$

4.5.7.4 Derivation of dominant wavelength. The dominant wavelength may be derived within an accuracy of ±1/2 nanometre by plotting the x, y chromaticity coordinates on the 1931 CIE Chromaticity Diagram and by drawing a line from the CIE Illuminant C point (6500°K color temperature point) through the x, y color point and intersecting the perimeter of the Chromaticity Diagram. The point where the line intersects the perimeter is the dominant wavelength.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-D-87157.

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6. NOTES

6.1 Notes. The notes specified in MIL-D-87157 are applicable to this specification.

6.2 Ordering data. The contract or purchase order may specify the inspection data (see MIL-D-87157).

6.3 Applied voltage limitation. A negative voltage greater than -0.5 V dc shall not be applied to any input or V_{CC} as the excessive voltage may damage the on-board integrated circuit.

6.4 Color binning. Color binning of the red LED displays shall not be required because the human eye is not able to discern small color differences in the red region where the radiated spectrum of these light-emitting diodes is located.

6.5 Substitutability. Displays covered by this specification are substitutable for the following commercial device types:

Device type	Commercial type
01	4N51
02	4N52
03	4N53
04	4N54

Custodian:
Army - ER

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5961-A832)

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TABLE I. Group A inspection.

Inspection	MIL-STD-883		LTPD	Symbol	Limits		Unit
	Method	Conditions			Min	Max	
Subgroup 1 1/			5				
Logic supply current	3005	V_{CC} = 5.5 V dc; (numeral 5 and decimal point lighted)		I_{CC1}		160	mA dc
Blanking input current "0" logic level 2/	---	V_{CC} = 5.5 V dc; V_{BL} = 0.8 V dc		I_{BL1}		20	µA dc
Blanking input current "1" logic level 2/	---	V_{CC} = 5.5 V dc; V_{BH} = 4.5 V dc		I_{BH1}		0.7	mA dc
Logic current "0" state	3009	V_{CC} = 5.5 V dc; V_{IL} = 0.4 V dc		I_{IL1}		-1.5	mA dc
Logic current "1" state	3010	V_{CC} = 5.5 V dc; V_{IH} = 2.4 V dc		I_{IH1}		50	µA dc
Enable current "0" state	---	V_{CC} = 5.5 V dc; V_{EL} = 0.4 V dc		I_{EL1}		-1.5	mA dc
Enable current "1" state	---	V_{CC} = 5.5 V dc; V_{EH} = 2.4 V dc		I_{EH1}		55	µA dc
Subgroup 2 1/		T_A = 100°C	7				
Logic supply current	3005	V_{CC} = 5.5 V dc; (numeral 5 and decimal point lighted)		I_{CC2}		150	mA dc
Blanking input current "0" logic level 2/	---	V_{CC} = 5.5 V dc; V_{BL} = 0.8 V dc		I_{BL2}		50	µA dc
Blanking input current "1" logic level 2/	---	V_{CC} = 5.5 V dc; V_{BH} = 4.5 V dc		I_{BH2}		0.6	mA dc
Logic current "0" state	3009	V_{CC} = 5.5 V dc; V_{IL} = 0.4 V dc		I_{IL2}		-1.4	mA dc
Logic current "1" state	3010	V_{CC} = 5.5 V dc; V_{IH} = 2.4 V dc		V_{IH2}		100	µA dc
Enable current "0" state	---	V_{CC} = 5.5 V dc; V_{EL} = 0.4 V dc		I_{EL2}		-1.4	mA dc
Enable current "1" state	---	V_{CC} = 5.5 V dc; V_{EH} = 2.4 V dc		I_{EH2}		130	µA dc
Subgroup 3 1/		T_A = -55°C	7				
Logic supply current 2/	3005	V_{CC} = 5.5 V dc; (numeral 5 and decimal point lighted)		I_{CC3}		170	mA dc
Blanking input current "0" logic level 2/	---	V_{CC} = 5.5 V dc; V_{BL} = 0.8 V dc		I_{BL3}		20	µA dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection	MIL-STD-883 Method	MIL-STD-883 Conditions	LTPD	Symbol	Limits Min	Limits Max	Unit
Subgroup 3 1/ -Continued		T_A = -55°C	7				
Blanking input current "1" logic level 2/	---	V_{CC} = 5.5 V dc; V_{BH} = 4.5 V dc		I_{BH3}		1.0	mA dc
Logic current "0" state	3009	V_{CC} = 5.5 V dc; V_{IL} = 0.4 V dc		I_{IL3}		-1.6	mA dc
Logic current "1" state	3010	V_{CC} = 5.5 V dc; V_{IH} = 2.4 V dc		I_{IH3}		20	µA dc
Enable current "0" state	---	V_{CC} = 5.5 V dc; V_{EL} = 0.4 V dc		I_{EL3}		-1.6	mA dc
Enable current "1" state	---	V_{CC} = 5.5 V dc; V_{EH} = 2.4 V dc		I_{EH3}		20	µA dc
Subgroups 4, 5, and 6							
Not applicable							
Subgroup 7			5				
Luminous intensity per LED average 3/	---	V_{CC} = 5.0 V dc		I_V	44	342	µcd
Functional test in accordance with truth table of figure 2	---	Logic low level = 0.8 V dc Logic high level = 2.0 V dc					
Subgroup 8			7				
External visual	2009						

1/ Does not apply to display type 03.

2/ Applies to 04 display type only.

3/ The following characters shall be displayed for the luminous intensity test:

Device types 01, 02, and 04 - - - - - - - - - - - - - - - - character "8"
Device type 03- character "+1" and decimal point

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TABLE II. Group B inspection.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			
Resistance to solvents	1022		4 devices/0 failures
Decap internal visual design verification 1/	2075		1 device/0 failure
Subgroup 2			LTPD = 15
Solderability	2026	Immersion within .062 inch of seating plane; solder bath temperature = 245 ±5°C	
Subgroup 3			LTPD = 15
Thermal shock (temperature cycling)	1051	Test condition B-1; $t_{transfer} \leq 5$ min.; $t_{extreme} \geq 10$ min.	
Moisture resistance	1021	See 4.5.3	
Hermetic seal	1071		
Fine leak		Test condition G or H; maximum leak rate = 5×10^{-8} atm cm^3/s	
Gross leak		Test condition A, C, E, or F	
Electrical/optical end points		Group A, subgroups 1 and 7	
Subgroup 4			LTPD = 10
Steady-state operation life	1027	T_A = 100°C; V_{CC} = 5.0 V dc; logic shall be cycled at 1 character per second (see 4.5.1)	
Electrical/optical end points		Group A, subgroups 1 and 7	
Subgroup 5			LTPD = 10
High-temperature (nonoperating) life (LTPD)	1032	T_A = 125°C (see 4.5.1)	
Electrical/optical end points		Group A, subgroups 1 and 7	

1/ Inspection may be performed through the clear glass window.

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TABLE III. Group C inspection.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1 1/			
Physical dimensions	2066		2 devices/0 failures
Subgroup 2			LTPD = 15
Lead integrity		See 4.5.2	
Hermetic seal	1071		
Fine leak		Test condition G or H; maximum leak rate = 5×10^{-8} atm cm^3/s	
Gross leak		Test condition A, C, E, or F	
Subgroup 3			LTPD = 15
Shock	2016	Nonoperating; 1500 G; 0.5 ms, 5 blows in each orientation X1, Y1, Z1.	
Vibration, variable frequency	2056		
Constant acceleration	2006	10,000 G minimum; Y1 orientation; 1 minute minimum	
Electrical/optical end points		Group A, subgroups 1 and 7	
Subgroup 4			LTPD = 15
Salt atmosphere	1041		
External visual		See 4.2.3	
Subgroup 5			LTPD = 20 (c = 0)
Bond strength	2037		
Subgroup 6			λ = 10
Steady-state operation life	1026	T_A = 100°C; V_{CC} = 5.0 V dc; logic shall be cycled at 1 character per second (see 4.5.1)	
Electrical/optical end points		Group A, subgroups 1 and 7	

1/ Measurements shall apply to external dimensions only.

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